

Zeke Mohammed

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Engineering Physics student with hands-on experience in PCB design, embedded systems, FPGA development, and computational research. Completing accelerated 3-year degree while managing concurrent research and teaching roles.

EDUCATION

Fordham University B.S. Engineering Physics, Electrical Concentration; Minor in Mathematics; GPA: 3.31 • Completing degree in 3 years while working 15–20 hrs/week in research and teaching roles • Sarah Colman Klemmer Research Fellow, Sigma Xi Honor Society, CSTEP Scholar • Coursework: E&M I/II, Quantum Mechanics, Circuits, Thermo & Statistical Mechanics, Linear Algebra, Multivariable Calculus, Experimental Techniques	Bronx, NY Expected May 2026
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TECHNICAL SKILLS

PCB & Embedded: KiCad (schematic, layout, Gerber, BOM), STM32, ESP32, Raspberry Pi, Arduino, C/C++, bare-metal programming FPGA: Xilinx Artix-7, Verilog, Vivado, pipelined architectures, fixed-point arithmetic Software: Python, MATLAB, Java, Git/GitHub, Linux, Flask, NumPy, pandas, matplotlib, LaTeX, SQL Protocols: CAN bus (ISO 11898), I2C, SPI, UART, PWM, OBD-II (ISO 15765-4) Hardware & Test: Oscilloscope, multimeter, signal analysis, SMD/through-hole soldering, PCB assembly & rework CAD: Fusion 360, SolidWorks, KiCad 3D viewer, Blender, 3D printing (FDM/ABS)

TECHNICAL PROJECTS

Yamaha ATV CAN Bus Sensor Node KiCad, STM32, CAN Bus, Fusion 360 • Designed production-ready 2-layer PCB (60×40 mm, ENIG) with STM32F103 and SN65HVD230 CAN transceiver; aggregates throttle position, coolant temp, wheel speed, and battery voltage into a single CAN 2.0B frame at 500 kbaud / 10 Hz • Full deliverable package: schematic, layout with solid ground plane and matched-length CAN differential pairs, Gerbers, interactive BOM (24 line items, \$15.10 prototype cost), and design rationale document • Designed injection-moldable ABS enclosure in Fusion 360: IP67-ready O-ring gasket groove, M6 ATV frame flanges, DB9 strain relief, countersunk lid screws, debossed Yamaha branding; <\$1/unit at production volume	March 2026
FPGA Real-Time Image Processing Accelerator Verilog, Vivado, Python • 7-stage pipelined convolution engine on Xilinx Artix-7 with systolic array and Q8.8 fixed-point arithmetic; 48–53× speedup over NumPy at 0.85 W. Three clock domains with CDC synchronizers; BRAM line buffer; Python-UART verification and VGA output	June 2025 – Present
Flight Controller + Power Distribution STM32, C, PID Control • Bare-metal STM32F405 firmware: 1 kHz cascaded PID loop with complementary filter sensor fusion (MPU6050 I2C/SPI); <2° RMS error, 655 μs loop time. Power system: 11.1 V 3S LiPo, four 20A BLHeli_S ESCs, overcurrent protection	November 2024 – May 2025
Vehicle Telemetry System ESP32, GPS, OBD-II, Bluetooth • Multi-protocol ESP32 system: GPS via UART (NEO-6M/TinyGPS++) and live ECU data over Bluetooth OBD-II (ISO 15765-4 CAN, 500 kbaud). Cross-validates dual speed sources on SSD1306 OLED with five cycling display screens	October 2024
Multi-Sensor DAQ System (M3) Python, Raspberry Pi, Flask • 12+ sensors across CAN/I2C/UART at 50 Hz; automated logging with Flask telemetry dashboard; <2% error over 50+ hrs	October 2023 – April 2024

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern Hyzon Motors • Characterized 200 kW hydrogen fuel cell stack via V-I polarization curve tests; Python post-processing to isolate ohmic, activation, and mass transport losses. HV validation (400–800 V DC): hipot, insulation resistance, continuity. BMS telemetry over CAN bus during fuel cell–battery hybrid integration testing	June 2024 – August 2024 Rochester, NY
Undergraduate Researcher, Lattice QCD Fordham University Physics Department • Built Python framework for SU(2) lattice gauge theory (8 ³ ×20 lattice): 81,920×81,920 sparse Wilson–Dirac operator, BiCGSTAB/GMRES solvers, 200+ Monte Carlo configs. Generalized SU(2)→SU(N) with 3× runtime improvement • Contributing editor for <i>High Energy Physics Essentials</i> (Wiley) by Dr. C. Aubin; presented at CSTEP Symposium	August 2024 – Present Bronx, NY
Teaching Assistant, Lab Instructor & Lab Manager Fordham University • TA: 30+ students, 120+ lab reports/month. Lab Manager: 50-workstation teaching lab, 99%+ uptime, 40% faster deployment	August 2023 – Present Bronx, NY
Freelance Electronics Technician Self-Employed • 50+ clients: console modifications, firmware flashing, custom PC builds, component-level SMD repair; 5.0/5.0 rating	January 2020 – Present Queens, NY

LEADERSHIP & ACTIVITIES

Fordham Rocket Propulsion Lab , Co-Founder (Sept 2024 – Present): Propulsion design, thrust calcs, trajectory modeling, Fusion 360 Physics & Math Tutor , Knack (June 2024 – Present): 5.0/5.0 across 50+ sessions Fordham Physics Society (Sept 2023 – Present)
